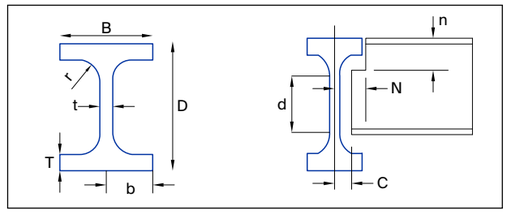
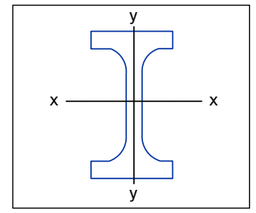
2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W8-8x6 1/2 (203x165)	24	35.7	201.4	165.0	10.1	6.2	10.2	160.8	45.5	8.17	25.9	5	89	20
	28	41.7	204.7	166.0	11.8	7.2	10.2	160.7	53.2	7.03	22.2	6	89	22
W8-8x8 (203x203)	31	46.1	203.2	203.6	11.0	7.2	10.2	160.8	58.7	9.25	22.3	6	108	21
	35	52.0	206.2	204.3	12.5	7.9	10.2	160.8	66.3	8.17	20.4	6	108	23
	40	60.0	209.6	205.8	14.2	9.4	10.2	160.8	76.4	7.25	17.1	7	108	24
	48	71.0	215.8	206.4	17.3	10.0	10.2	160.8	90.4	5.97	16.1	7	108	28
	58	86.1	222.2	209.1	20.5	12.7	10.2	160.8	110	5.10	12.7	8	108	31
	67	99.7	228.6	210.3	23.8	14.5	10.2	160.7	127	4.43	11.1	9	108	34
W10-10x4 (254x102)	12	17.9	250.7	100.6	5.3	4.8	7.6	224.8	22.8	9.44	46.5	4	58	13
	15	22.0	254.0	101.6	6.8	5.7	7.6	225.2	28.0	7.47	39.5	5	58	14
	17	25.2	257.2	101.9	8.4	6.0	7.6	225.2	32.0	6.07	37.5	5	58	16
	19	28.3	260.4	102.2	10.0	6.3	7.6	225.2	36.1	5.11	35.7	5	58	18
W10-10x5 3/4 (254x146)	21	31.1	251.4	146.1	8.6	6.0	7.6	219.0	39.7	8.49	36.5	5	80	16
	22	32.7	258.3	146.0	9.1	6.1	7.6	224.8	41.8	7.99	36.8	5	80	17
	25	37.0	256.0	146.4	10.9	6.3	7.6	219.0	47.2	6.72	34.8	5	80	19
	26	38.7	262.4	146.6	11.1	6.6	7.6	225.0	48.9	6.60	34.1	5	80	19
	29	43.0	259.6	147.3	12.7	7.2	7.6	219.0	54.8	5.80	30.4	6	80	20
	30	44.6	265.9	147.6	12.9	7.6	7.6	224.9	56.9	5.72	29.5	6	80	21
W10-10x8 (254x203)	33	49.1	247.1	202.2	11.0	7.4	12.7	199.7	62.5	9.19	27.1	6	107	24
	39	58.0	252.0	202.8	13.4	8.0	12.7	199.8	73.8	7.57	25.0	6	107	26
	42	62.5	249.4	207.0	12.2	12.2	12.7	199.6	79.3	8.49	16.4	8	107	25
	45	67.0	256.5	203.7	15.7	8.9	12.7	199.7	85.4	6.49	22.5	6	107	28
	57	84.8	258.1	211.3	16.5	16.5	12.7	199.7	108	6.40	12.1	10	107	29
W10-10x10 (254x254)	49	73.1	254.1	254.6	14.2	8.6	12.7	200.3	93.1	8.96	23.3	6	133	27
	54	80.4	256.3	254.8	15.6	9.4	12.7	199.7	102	8.17	21.2	7	133	28
	60	88.9	260.3	256.3	17.3	10.3	12.7	200.3	113	7.41	19.4	7	133	30
	68	101.2	264.2	257.3	19.6	11.9	12.7	199.7	129	6.58	16.7	8	133	32
	72	107.1	266.7	258.8	20.5	12.8	12.7	200.3	136	6.31	15.6	8	133	33
	77	114.6	269.2	258.8	22.1	13.5	12.7	199.6	146	5.86	14.8	9	133	35
	88	131	275.3	260.7	25.2	15.4	12.7	199.6	167	5.18	13.0	10	133	38
	89	132	276.3	261.3	25.3	15.3	12.7	200.3	168	5.16	13.1	10	133	38
	100	148.8	281.9	262.6	28.5	17.3	12.7	199.6	190	4.62	11.6	11	133	41
	112	167.1	289.1	265.2	31.7	19.2	12.7	200.3	213	4.18	10.4	12	133	44

Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.03	3430	757	8.68	4.08	341	91.7	378	140	0.875	19.3	0.069	14.3
1.04	4077	901	8.76	4.12	398	109	445	165	0.875	16.8	0.084	22.4
1.19	4568	1548	8.82	5.13	450	152	497	231	0.847	17.7	0.143	22.2
1.20	5259	1778	8.91	5.18	510	174	567	264	0.848	15.8	0.167	31.8
1.21	6125	2065	8.96	5.20	584	201	656	305	0.846	14.1	0.197	47.2
1.22	7618	2537	9.18	5.30	706	246	799	374	0.853	11.9	0.250	80.2
1.24	9449	3128	9.28	5.34	850	299	977	456	0.850	10.2	0.318	137
1.25	11310	3687	9.44	5.39	990	351	1150	536	0.852	9.0	0.387	211
0.881	2241	90.8	9.91	1.99	179	18.0	207	28.6	0.851	44.0	0.014	2.28
0.890	2841	119	10.1	2.06	224	23.5	259	37.3	0.856	36.4	0.018	4.15
0.897	3415	149	10.3	2.15	266	29.2	306	46.0	0.866	31.5	0.023	6.42
0.904	4005	179	10.5	2.22	308	34.9	353	54.8	0.874	27.5	0.028	9.57
1.06	4413	448	10.5	3.36	351	61.3	393	94.1	0.880	29.6	0.066	8.55
1.08	4917	475	10.8	3.37	381	65.0	426	100	0.882	28.9	0.074	9.96
1.07	5537	571	10.8	3.48	433	78.0	483	119	0.890	24.3	0.086	15.3
1.08	5974	584	11.1	3.45	455	79.6	510	122	0.888	24.5	0.092	16.5
1.08	6544	677	10.9	3.52	504	92.0	566	141	0.891	21.2	0.103	23.9
1.09	7048	692	11.1	3.49	530	93.8	597	144	0.888	21.3	0.111	25.6
1.27	7071	1517	10.6	4.93	572	150	634	229	0.873	21.5	0.211	24.0
1.28	8672	1865	10.8	5.03	688	184	765	280	0.878	18.3	0.265	40.2
1.25	8430	1807	10.3	4.77	676	175	768	271	0.851	17.9	0.254	44.5
1.29	10300	2214	11.0	5.09	803	217	898	331	0.880	15.9	0.321	62.3
1.31	11930	2606	10.5	4.91	925	247	1067	385	0.849	13.6	0.380	109
1.49	11410	3908	11.1	6.48	898	307	992	465	0.849	17.3	0.562	57.6
1.49	12590	4304	11.1	6.49	983	338	1091	512	0.849	15.8	0.623	75.6
1.50	14270	4858	11.2	6.55	1096	379	1224	575	0.850	14.5	0.717	102
1.51	16390	5558	11.3	6.57	1241	432	1398	657	0.849	12.9	0.832	148
1.52	17510	5928	11.3	6.59	1313	458	1484	697	0.848	12.4	0.898	172
1.52	18950	6391	11.4	6.61	1408	494	1599	752	0.849	11.6	0.976	213
1.54	22210	7436	11.5	6.67	1613	570	1850	869	0.850	10.3	1.16	314
1.55	22530	7531	11.6	6.69	1631	576	1869	878	0.850	10.3	1.19	319
1.56	25900	8598	11.7	6.73	1838	655	2127	999	0.851	9.3	1.38	452
1.58	30000	9870	11.9	6.81	2075	744	2424	1137	0.851	8.5	1.63	626